

Blind Calibration and DOA Estimation With Uniform Circular Arrays in the Presence of Mutual Coupling

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Abstract—In smart antenna systems, mutual coupling between elements can significantly degrade the performance of array processing algorithms. Based on the special structure of coupling matrix, this letter presents a subspace-based blind calibration method for uniform circular arrays. The method uses the incident signals to carry out both direction of arrival (DOA) estimation and array calibration simultaneously. In addition, the parameter identifiability condition is also provided. Finally, representative computer simulation results are given to demonstrate the effectiveness and behavior of the proposed method.

Index Terms—Array mutual coupling, array signal processing, blind calibration, direction of arrival (DOA) estimation.

I. INTRODUCTION

WHILE the study of smart antennas in radar and wireless communications has been an active research topic for several decades, it has shown that most typical adaptive beamforming and high resolution direction-finding (DF) algorithms, are sensitive to array mutual coupling, and such effect needs to be removed to achieve theoretical high performance in an actual system [1], [2]. In the literature, there are a number of works that address this issue. Generally, the methods may be categorized into two classes: deterministic methods and blind methods. The deterministic methods are offline array calibration techniques, which have been widely studied in the previous articles. In [1], the authors considered the antenna array as a linear bilateral network, and derived the relationship between the element output voltages and the open circuit voltages, through using the mutual coupling matrix as a transform matrix. The work of [1] is the first published effort analyzing and compensating for the effect of mutual coupling on adaptive arrays. The method was later used in [3] to eliminate the mutual coupling effect on direction of arrival (DOA) estimation algorithm. Based on these works, many calibration methods have been proposed by different researchers. For example, in [4] and [5], the author employed a set of time-invariant calibration sources with precisely known DOAs to estimate the mutual coupling matrix, and eliminated the coupling effect with matrix compensation

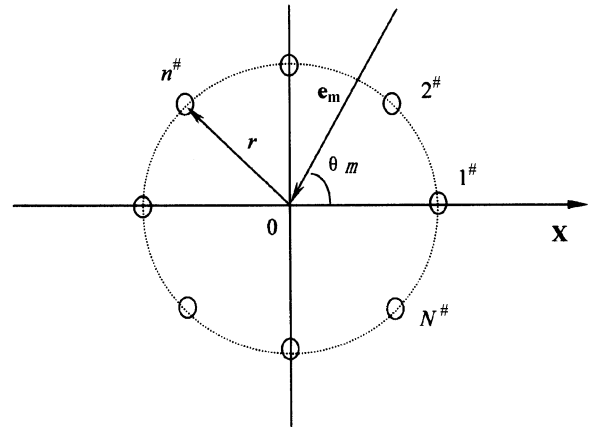


Fig. 1. Planar uniform circular array geometry.

as described in [3]. In [6], the authors proposed another subspace-based method, where the number of calibration sources can be reduced to one, and its DOA may be unknown. However, the major drawback of these algorithms is that sometimes the additional calibration sources are not available in practice. Meanwhile, in [2], [7] and [8], the authors tried to analyze and compensate the mutual coupling effect using the computational electromagnetic solvers, but they ignored the dependence of coupling matrix on the environment. On the other hand, the blind methods are online array calibration techniques [9]–[11]. They are very appealing because they can estimate the needed parameters and eliminate the coupling effects simultaneously. However, most of the algorithms show a computational complexity and sometimes no unique solutions, so further studies need to be done in this area.

Compared with uniform linear array (ULA), uniform circular array (UCA) has the advantages of more compact structure and inherent symmetry in azimuth, thus it is a highly desirable choice for full peripheral coverage and wide bandwidth application. However, mutual coupling effect of UCA is much stronger than that of ULA, so we focus on uniform circular arrays and present a blind calibration method in this letter. The proposed method uses the special structure of mutual coupling matrix and the subspace principle, and provides an accurate DOA estimation as well as array calibration.

II. PROBLEM FORMULATION

Consider a circular array of N elements impinged by M narrow-band signals from far field, as shown in Fig. 1. With the presence of array mutual coupling, the signal at the output of an individual element has several constituents: a dominant one due

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to the direct incident radiation, and several lesser ones due to the reflected radiation from its surrounding elements. Hence, the actual array output vector $\mathbf{X}(t) = [x_1(t), x_2(t), \dots, x_N(t)]^T$, where T denotes transpose, can be described by

$$\mathbf{X}(t) = \mathbf{C}\mathbf{A}\mathbf{S}(t) + \mathbf{n}(t) \quad (1)$$

where $\mathbf{S}(t)(M \times 1)$ and $\mathbf{n}(t)(N \times 1)$ represent the signal vector and the noise vector respectively. The m th column of $\mathbf{A}$, which is represented by $\mathbf{a}(\theta_m)$, is the ideal array steering vector for the m th signal source, and is given by

$$\mathbf{a}(\theta_m) = \begin{bmatrix} \exp(j\beta r \cos(\theta_m)) \\ \exp(j\beta r \cos(\theta_m - 2\pi/N)) \\ \vdots \\ \exp(j\beta r \cos(\theta_m - 2\pi(N-1)/N)) \end{bmatrix} \quad (2)$$

where r is the radius of the circular array, β is the wavenumber, namely, $\beta = 2\pi/\lambda$. With the assumption of single-mode elements, the mutual coupling matrix (MCM) $\mathbf{C}$ is generally considered to be independent of angle [12]. According to the fundamental electromagnetic and circuit theory, it can be written as [13], [14]

$$\mathbf{C} = (Z_A + Z_L)(\mathbf{Z} + Z_L\mathbf{I}_N)^{-1} \quad (3)$$

where Z_A is the element impedance, Z_L is the load impedance, $\mathbf{I}_N$ is the identity matrix and $\mathbf{Z}$ is the mutual impedance matrix. With the classical induced electromotive force (EMF) method, the pq th component of mutual impedance matrix of dipoles with equal length can be expressed as [14]–[16]

$$[Z]_{pq} = \begin{cases} 30[0.5772 + \ln(2\beta l) - C_i(2\beta l) \\ \quad + j30S_i(2\beta l)] & p = q \\ R_{pq} + jX_{pq} & p \neq q \end{cases} \quad (4)$$

where

$$\begin{aligned} R_{pq} &= 30 \{ \sin(\beta l) [-S_i(u_0) + S_i(v_0) + 2S_i(u_1) - 2S_i(v_1) \\ &\quad + \cos(\beta l) [C_i(u_0) + C_i(v_0) - 2C_i(u_1) - 2C_i(v_1) \\ &\quad + 2C_i(\beta d_{pq})] - [2C_i(u_1) + 2C_i(v_1) - 4C_i(\beta d_{pq})] \} \\ X_{pq} &= 30 \{ \sin(\beta l) [-C_i(u_0) + C_i(v_0) + 2C_i(u_1) - 2C_i(v_1) \\ &\quad + \cos(\beta l) [-S_i(u_0) - S_i(v_0) + 2S_i(u_1) + 2S_i(v_1) \\ &\quad - 2S_i(\beta d_{pq})] + [2S_i(u_1) + 2S_i(v_1) - 4S_i(\beta d_{pq})] \}. \end{aligned}$$

In the previous equations, we use the following notation:

$$\begin{aligned} u_0 &= \beta \left(\sqrt{d_{pq}^2 + l^2} - l \right) & v_0 &= \beta \left(\sqrt{d_{pq}^2 + l^2} + l \right) \\ u_1 &= \beta \left(\sqrt{d_{pq}^2 + 0.25l^2} - l/2 \right) \\ v_1 &= \beta \left(\sqrt{d_{pq}^2 + 0.25l^2} + l/2 \right) \\ C_i(u) &= \int_{-\infty}^u \frac{\cos(x)}{x} dx & S_i(u) &= \int_0^u \frac{\sin(x)}{x} dx \end{aligned}$$

where d_{pq} is the distance between dipoles p and q , l is the length of dipole. Without loss of generality, we suppose that the signal sources are uncorrelated and the noises are independent with

zero mean and σ_n^2 variance, the spatial covariance matrix of $\mathbf{X}(t)$ can be written as

$$\mathbf{R} = E[\mathbf{X}(t)\mathbf{X}^H(t)] = \mathbf{C}\mathbf{A}\mathbf{R}_{SS}(\mathbf{C}\mathbf{A})^H + \sigma_n^2\mathbf{I} \quad (5)$$

where $\mathbf{R}_{SS} = E[\mathbf{S}(t)\mathbf{S}^H(t)]$, E and H denote expectation and complex conjugate transpose respectively. Decomposing the spatial covariance matrix $\mathbf{R}$, we denote the M eigenvectors corresponding to the M largest eigenvalues and the rest as $\mathbf{E}_S$ and $\mathbf{E}_N$, respectively. According to the subspace principle, the following relation can be obtained:

$$\text{span}\{\mathbf{C}\mathbf{A}(\theta)\} = \text{span}\{\mathbf{E}_S\} \perp \text{span}\{\mathbf{E}_N\}. \quad (6)$$

Thus, the spatial spectrum function of standard multiple-signal-classification (MUSIC) algorithm should be modified as

$$P_{MU}(\theta) = \frac{1}{\|\mathbf{E}_N^H \mathbf{C}\mathbf{a}(\theta)\|^2} \quad (7)$$

where $\|\bullet\|$ denotes 2-norm. The DOAs of M signals, namely, $\{\theta_m\}_{m=1}^M$, can be estimated by searching the spectrum peaks as a function of angle θ .

III. NEW BLIND CALIBRATION METHOD

Since mutual coupling coefficients between two elements are inversely related to their distances, the coupling matrix of a good uniform circular array can be expressed as a banded complex symmetric circular Toeplitz matrix. Therefore, we can do a certain kind of matrix–vector transform and have

$$\mathbf{C}\mathbf{a}(\theta) = \mathbf{T}(\theta)\mathbf{c}. \quad (8)$$

As we know, a fixed complex difference among all of the components of array steering vector does not affect the performance of DOA estimation, thus the main diagonal of mutual coupling matrix can be normalized to unity, and the components of the vector $\mathbf{c}$ are given by

$$c_1 = 1 \quad \text{and} \quad c_i = C_{1i}(i = 2, 3, \dots, K) \quad (9)$$

where $K = \lfloor N/2 + 1 \rfloor$, and $\lfloor s \rfloor$ denotes the maximum integer less than or equals to s . The $N \times K$ matrix $\mathbf{T}(\theta)$, which is named as transform matrix, is the first K column of the sum of the four following matrices, that is:

$$\mathbf{T}_1 + \mathbf{T}_2 + \mathbf{T}_3 + \mathbf{T}_4 = [\mathbf{T}; \tilde{\mathbf{T}}] \quad (10)$$

and the pq th components of $\mathbf{T}_i$ ($i = 1, 2, 3, 4$), namely, $[T_i]_{pq}$, are given by the following equations, respectively:

$$[T_1]_{pq} = \begin{cases} \mathbf{a}(\theta)_{p+q-1}, & \text{for } p+q \leq N+1 \\ 0, & \text{otherwise} \end{cases} \quad (11a)$$

$$[T_2]_{pq} = \begin{cases} \mathbf{a}(\theta)_{p-q+1}, & \text{for } p \geq q \geq 2 \\ 0, & \text{otherwise} \end{cases} \quad (11b)$$

$$[T_3]_{pq} = \begin{cases} \mathbf{a}(\theta)_{N+p-q+1}, & \text{for } p < q \leq l \\ 0, & \text{otherwise} \end{cases} \quad (11c)$$

$$[T_4]_{pq} = \begin{cases} \mathbf{a}(\theta)_{p+q-N-1}, & \text{for } 2 \leq q \leq l, \\ & p+q \geq N+2 \\ 0, & \text{otherwise} \end{cases} \quad (11d)$$

where $l = \lfloor (N + 1)/2 \rfloor$. Consider that the mutual coupling coefficients between two elements, which are far enough from each other, can be approximated as zeros, we suppose $c_{L+1} = c_{L+2} = \dots = c_K$ and obtain

$$\mathbf{C}\mathbf{a}(\theta) = \mathbf{T}(\theta)\mathbf{c} = \begin{bmatrix} \mathbf{D}(\theta) & \mathbf{E}(\theta) \end{bmatrix} \begin{bmatrix} \mathbf{g} \\ \mathbf{e} \end{bmatrix} = \mathbf{D}(\theta)\mathbf{g} \quad (12)$$

where $\mathbf{D}(\theta)$ and $\mathbf{E}(\theta)$ are $N \times L$ and $N \times (K - L)$ matrices respectively, and

$$\mathbf{g} = [1, c_2, \dots, c_L]^T \quad (13)$$

$$\mathbf{e} = [c_{L+1}, c_{L+2}, \dots, c_K]^T = [0, 0, \dots, 0]^T. \quad (14)$$

Therefore, the joint estimation of DOAs and coupling coefficients can be expressed as the following minimization problem:

$$(\theta_i, \mathbf{g}) = \arg \min_{(\theta_i, \mathbf{C})} \|\mathbf{E}_N^H \mathbf{C}\mathbf{a}(\theta)\|^2 = \arg \min_{(\theta_i, \mathbf{g})} \mathbf{g}^H \mathbf{Q}(\theta) \mathbf{g} \quad (15)$$

where $\mathbf{Q}(\theta)$ ($L \times L$) is given by

$$\mathbf{Q}(\theta) = \mathbf{D}^H(\theta) \mathbf{E}_N \mathbf{E}_N^H \mathbf{D}(\theta). \quad (16)$$

Let $\mathbf{w} = [1, 0, \dots, 0]^T$, we get the constraint

$$\mathbf{g}^H \mathbf{w} \equiv 1. \quad (17)$$

Equations (17) and (15) constitute a constraint quadratic minimization problem. The result is often obtained with the method of Lagrange multipliers, and is given by

$$\mathbf{g} = \frac{\mathbf{Q}^{-1}(\theta) \mathbf{w}}{\mathbf{w}^H \mathbf{Q}^{-1}(\theta) \mathbf{w}}. \quad (18)$$

Substituting (18) into (15) and (7), the modified MUSIC spectrum function can be written as

$$P_{MU}(\theta) = \mathbf{w}^H \mathbf{Q}^{-1}(\theta) \mathbf{w}. \quad (19)$$

Finally, we discuss the parameter identifiability condition of the algorithm. To get the unique solution of (19), $\mathbf{Q}(\theta)$ must be invertible, that is

$$\text{rank}(\mathbf{Q}(\theta)) = L. \quad (20)$$

Furthermore, the rank of $\mathbf{E}_N \mathbf{E}_N^H$ is $N - M$, thus

$$\text{rank}(\mathbf{Q}(\theta)) \leq N - M. \quad (21)$$

So the parameter identifiability condition of the algorithm is

$$L \leq N - M. \quad (22)$$

The same result has also been obtained by Fieller *et al.* [10] through different method.

IV. SIMULATION EXPERIMENTS

In this section, we carry out some representative simulation experiments to demonstrate the validity of the proposed calibration method. In all these experiments, we suppose that the central frequency is 2.4 GHz and study a 7-element uniform circular

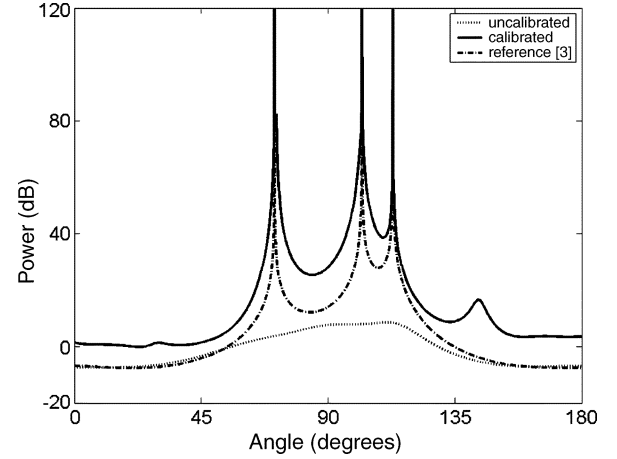


Fig. 2. MUSIC spatial spectra for three signals at 2.4 GHz.

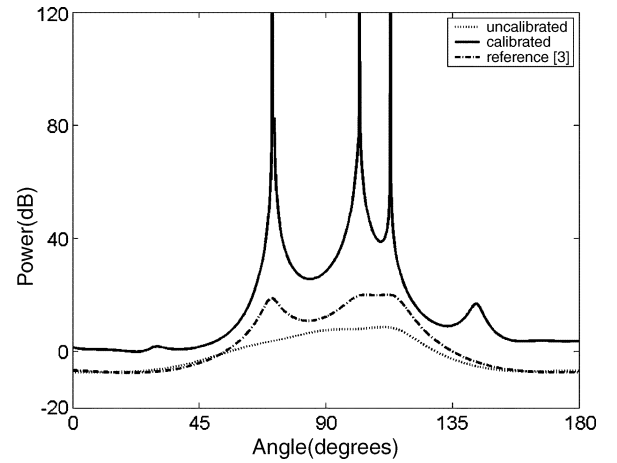


Fig. 3. Comparison of spatial spectra for 7-element UCA at 2.3 GHz.

dipole array with the radius of 72 mm. We also suppose that the load impedance Z_L is chosen as the complex conjugate of Z_A to obtain an impedance match for maximum power transfer and the number of snapshots is 512.

First of all, consider three uncorrelated sources, each with a SNR of 20 dB, imping on the array from 71° , 102° and 113° , respectively. We set the number of coupling coefficients to 4, namely, $L = 4$, and give the MUSIC spatial spectra in Fig. 2, where the dotted curve, the solid curve and the dash-dot curve represent the uncalibrated result, the calibrated result using the proposed blind method and the calibrated result using the method of [3] respectively. As we can see, the performance degradation due to array mutual coupling is rather severe and the uncalibrated curve shows only one broad peak. However, the two calibrated spectra both have three sharp peaks on the correct locations. Hence, they can obtain the correct DOAs of incident signals.

Next, to further observe the effectiveness and robustness of the new method, we change the work frequency and do the same experiments. When the operating frequency is reduced to 2.3 GHz, the simulation results are shown in Fig. 3. It can be observed that the uncalibrated spectrum is too poor to provide

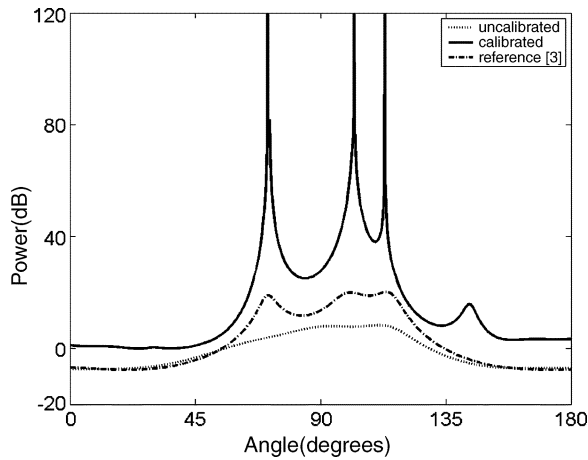


Fig. 4. Comparison of spatial spectra for 7-element UCA at 2.5 GHz.

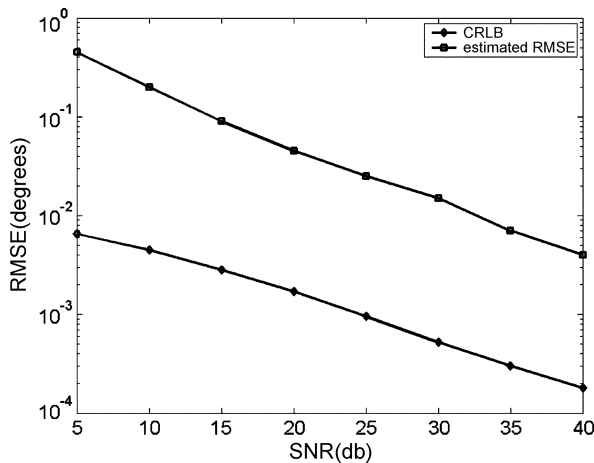


Fig. 5. Estimated RMSE and the corresponding CRLB versus SNR.

DOA estimation, while the calibrated one via the new method can resolve the three signals accurately. For comparison, we employ the method of [3], where the coupling matrix used to modify the search vector is identical to the first experiment, and plot another calibrated result in the same figure. We find that the spectrum no longer has sharp peaks to indicate the signals from 102° and 113° . The scenario can be attributed to the error between the actual coupling matrix and the compensation matrix. This implies that the method of [3] is sensitive to the change of operating environment. On the other hand, we increase the operating frequency to 2.5 GHz and give the simulation results in Fig. 4. It is not surprised that while the method of [3] can not resolve the two incident signals from 102° and 113° , the proposed method can still provide the correct DOA estimation of the three signals. The uncalibrated result is again the poorest.

In the third experiment, we take the 7-element UCA described previously, and use the results of the 50 Monte Carlo experiments to compute the root-mean-square error (RMSE) of the signal from 71° at 2.4 GHz. The estimated RMSE versus SNR and the corresponding Cramer–Rao lower bound (CRLB) are both plotted in Fig. 5. The figure clearly demonstrates the statistical efficiency of the proposed algorithm.

Finally, we change the experiment parameters and repeat this type of simulations. We find that the proposed method is effective in all cases. More simulation results can not be given due to the limitation of space.

V. CONCLUSION

In this letter, we consider uniform circular arrays and present a blind calibration method for mutual coupling between elements. The proposed method uses the special structure of mutual coupling matrix and the subspace principle, and provides an accurate DOA estimation as well as array calibration. Computer simulation results demonstrate that the method is correct and effective.

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